AS^{3}ITransUNet: Spatial–Spectral Interactive Transformer U-Net With Alternating Sampling for Hyperspectral Image Super-Resolution

Qin Xu, Shiji Liu, Jiahui Wang, Bo Jiang, and Jin Tang

***Abstract*— Single hyperspectral image (HSI) super-resolution (SR) is an important topic in the remote-sensing field. However, existing HSI SR methods mainly use the feed-forward upsampling technique and convolutional neural network (CNN) to learn the feature representation, failing to learn the complex mapping relationship between low-resolution (LR) and high-resolution (HR) and long-range joint spectral and spatial features. To address this issue, in this article, we propose the spatial–spectral interactive transformer U-Net with alternating sampling (AS^{3}ITransUNet) for the HSI SR task. In this method, to mitigate the computational burden resulting from the high spectral dimension of the HSI, a group reconstruction strategy is adopted. To effectively explore the hierarchical features of the HSI, the U-Net with alternating upsampling and downsampling is designed that allocates the task of learning the complex mapping relationship to each stage of the U-Net. To fully extract the spatial–spectral features of the HSI, we propose the spatial–spectral interactive transformer (SSIT) block and integrate it into the encoder and the decoder of U-Net. The SSIT block contains a cross-branch bidirectional interaction module, which further captures the complementary information between spatial and spectral dimensions. Moreover, multistage complementary information learning (MSCIL) is proposed to capture the complementary information in the adjacent HSI groups for recovering the absent details in the current HSI group. The experiments on the three benchmark datasets demonstrate that the proposed AS^{3}ITransUNet can effectively improve the spatial resolution and preserve the spectral information at different scales. Models and codes are available at https://github.com/liushiji666/AS3-ITransUNet.**

***Index Terms*— Convolutional neural network (CNN), feature extraction, hyperspectral image (HSI) super-resolution (SR), transformer.**

I. Introduction

HYPERSPECTRAL image (HSI), which records the solar radiation signals reflected from the ground object, contains a large number of spectral bands from the visible to near-infrared wavelength. Because of the rich spectral and collaborative spatial information, hyperspectral remote-sensing science has derived various applications, such as clustering [1], classification [2], [3], and target detection [4]. However, the production and storage limitations of hyperspectral imaging equipment often result in imbalanced spatial and spectral resolutions. Compared with multispectral images or natural images, HSIs have higher spectral resolution with lower spatial resolution, which usually limits their applications in real-world scenes. It is more feasible to develop software technology than to upgrade the hardware equipment for improving the spatial resolution of HSIs.

HSI super-resolution (SR) aims to address this issue by producing the high-resolution (HR) HSI from the low-resolution (LR) HSI. HSI SR methods are classified into two categories, that is, fusion-based HSI SR method and single HSI SR method. Fusion-based HSI SR methods require the auxiliary images with spatial information of HR to enhance the spatial resolution [5], [6], [7], [8], [9], [10], [11]. However, this kind of method assumes that the auxiliary image should be well aligned with the input LR HSI. However, it is usually difficult to obtain accurate coregistration pairs in practice. In contrast, the single HSI SR methods only use the LR HSI to infer the corresponding HR HSI without any additional information. We focus on the single HSI SR methods in this article.

For the natural image SR problem, many methods have been proposed in recent years [12], [13], [14], [15], [16], [17], [18]. However, these image SR methods cannot be directly adopted for the HSI SR problem. The main possible reasons are twofold. First, HSIs consist of hundreds or thousands of spectral bands, which makes the feature representation of HSIs more difficult than natural images. Second, there are high correlations between the spectral bands of HSIs that should be considered. If the natural image SR method is directly applied to the HSI SR, it will lead to spectral distortion.

The early single HSI SR methods, including the sparse representation [19], [20] low rank [21], [22], and self-similarity [23], [24] are based on the manually generating complex priors. However, these traditional machine-learning methods are easily affected by the boundary effects and may lose some high-frequency details. Compared with the traditional machine-learning methods, deep neural networks (DNNs) can automatically learn the transformation between the LR and HR images. For example, Li et al. [25] used a recursive structure and 2-D convolution, which combines both the spectral angle mapper (SAM) and mean square error (MSE), to design the

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The authors are with the Key Laboratory of Intelligent Computing and Signal Processing of the Ministry of Education, School of Computer Science and Technology, Anhui University, Hefei 230601, China (e-mail: xuqin@ahu.edu.cn; e21201107@stu.ahu.edu.cn; e21301179@stu.ahu.edu.cn; jiangbo@ahu.edu.cn; tj@ahu.edu.cn).

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grouped deep recursive residual network (GDRRN) for the HSI SR problem. Jiang et al. [26] proposed to combine the spectral and spatial characteristics of HSIs to effectively improve the spatial resolution and retain the spectral information. In addition, Jiang et al. [27] proposed the use of a feedback network to mine the complementary information between bands of HSIs. However, 2-D convolution cannot mine the joint spectral and spatial features. Mei et al. [28] proposed to employ a 3-D fully convolutional neural network (3D-FCNN) for HSI SR to extract spatial and spectral features. Hu et al. [29] proposed the 3-D multiscale feature fusion aggregation network for HSI SR, which can effectively utilize the multiscale features. Also, the HSI SR methods [30], [31], [32] based on hybrid 2-D and 3-D convolutions were proposed. When the spectral information is extracted, more 2-D convolutions help to improve the spatial learning ability. However, the convolutional neural network (CNN)-based HSI SR methods generally fail to effectively extract the global spatial correlation of HSI data due to the local property of the convolution kernel.

Moreover, inspired by the generative adversarial network (GAN), Li et al. [33] proposed 3-D GAN for HSI SR, which uses GAN to preserve more texture details and uses 3-D convolution to explore the spectral correlation. Jiang et al. [34] proposed 2-D–1-D GAN frameworks for HSI SR. It uses two subnetworks for spatial and spectral dimensions to reduce the computational complexity of the network. Shi et al. [35] proposed a novel GAN model to improve the spatial–spectral reconstruction ability of HSIs and alleviate the issue of collapse. Recently, Jia et al. [36] proposed the diffused CNN for HSI SR, which designs a diffused image enhancement module to enlarge the receptive field. However, the method still utilizes the 2-D CNN to construct the SR HSI, which cannot effectively explore the spectral information.

The transformer model that can model the global correlation of image patches by self-attention was introduced for HSI SR [37], [38]. In [37], a two-branch network composed of the transformer and 3-D CNN was proposed for HSI SR, where the transformer branch is used to model the long-range dependencies and obtain the global spatial features of HSIs, and the 3-D CNN branch is used to extract the local features and retain the spectral correlation of HSIs. However, the 3-D CNN branch cannot capture the long-distance spectral dependencies of HSIs. In [38], the dual-self-attention swin transformer (DSST) was proposed for HSI SR, where the DSST module connects the spatial self-attention and spectral self-attention successively to extract the global spatial information and reduce the distortion of spectral information. However, the spectral attention and spatial attention in this method are separate, which limits the learning ability of the model.

Existing HSI SR methods usually use one or more upsampling layers to improve the resolution and predict the SR image in a feed-forward upsampling manner. However, it is difficult to learn the transformation of HSIs from LR to HR through a feed-forward upsampling manner because of the limited training samples. Additionally, to reduce memory consumption, some works adopt the grouping strategy that groups the input LR HSI into several groups along the spectral dimension. We refer to this kind of method as the group-based HSI SR methods [26], [27], [31], [39]. In these methods, considering the high similarity between adjacent spectral bands, complementary information of adjacent HSI groups is obtained by concatenating the adjacent HSI groups or capturing the differences between adjacent HSI groups in one stage. However, one limitation is that the feature will gradually fade away as the network deepens, which leads to the undesired effect that the complementary information from the adjacent groups is not well-exploited.

To address the aforementioned problems, in this article, we propose a novel spatial–spectral interactive transformer U-Net with alternating sampling (AS^3ITransUNet) for HSI SR. The proposed AS^3ITransUNet adopts the U-Net architecture and uses the partial HSI bands as the input. The encoder and the decoder of U-Net include several spatial–spectral interactive transformer (SSIT) blocks, where the alternating upsampling and downsampling operations are adopted. The SSIT is proposed to simultaneously extract the global spatial correlation and long-range spectral similarity of HSIs and provide supplementary clues in both spatial and spectral dimensions. Moreover, multistage complementary information learning is developed to learn the complementary information of the current and adjacent group features to supplement the SR of the current band's group at different depths of the decoder. For the loss computation, the final output result and the HR feature map of the intermediate process are supervised by the ground truth, respectively, to improve the feature representation of the network.

Overall, the main contributions of this work are summarized as follows.

1) We propose the AS^3ITransUNet for HSI SR. The proposed AS^3ITransUNet effectively combines the advantages of alternating up- and downsampling and U-Net architecture, thus enhancing the learning ability and flexibility of the traditional encoder–decoder architecture for HSI SR.
2) We propose the SSIT block that includes the bidirectional interaction of the spectral branch and the spatial branch. Different from the existing transformer models for HSI SR, both spectral and spatial branches in the SSIT block are learned cooperatively, encouraging information communication between the two branches.
3) We propose multistage complementary information learning to allow the aggregation of LR and HR HSI data features at different stages of the network. This can effectively fuse the contextual and complementary information between the adjacent HSI bands.

II. Related Works

The proposed AS^3ITransUNet leverages the transformer architecture and alternating upsampling and downsampling strategies. Previous works have generally been studied from two aspects, that is, the sampling strategy in the SR and vision transformer.

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A. Sampling Strategies in Super-Resolution

1) Predefined Upsampling: The predefined upsampling model was first proposed by Dong et al. [14]. It is the first end-to-end CNN SR model, named SRCNN. This model uses some interpolation algorithms to generate medium-resolution (MR) images and then uses a simple convolutional layer to learn the nonlinear mapping between MR and HR images. Subsequently, some works proposed the use of residual learning [15], [40] and recursive layers [41] to improve the SR performance of the network. However, these methods may generate some noises from MR images.

2) Single Upsampling: Single upsampling is the most commonly used method in image SR methods. This method was proposed by Dong et al. [42], who introduced a deconvolution layer at the end of the network and then directly learned the mapping from the LR image to the HR image. These methods can effectively improve the spatial resolution of images. Most of the existing natural image SR and HSI SR methods use this upsampling strategy [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37].

3) Progressive Upsampling: Lai et al. [43] proposed progressive upsampling for natural image SR via progressive reconstruction, that is, stacking multiple single upsampling networks. Inspired by SR of natural images, Jiang et al. [26] proposed to apply the progressive upsampling strategy to the single HSI SR and achieved good results. However, due to its progressive sampling strategy, the model cannot be applied to smaller scale factors (such as $\times 2$ and $\times 3$).

4) Iterative Upsampling and Downsampling: Iterative upsampling and downsampling techniques are proposed by deep backprojection networks (DBPN) [44], which are designed for SR of natural images. Specifically, it is a network model that alternately performs upsampling and downsampling, which enables the network to retain HR components by learning various sampling operators.

It can be seen that the process of image SR mainly learns the mapping between LR and HR images. Due to the high-dimensional information of HSIs, the feed-forward one-way upsampling manner may not be able to effectively capture the complex mapping relationships required for high-quality SR. Therefore, we use the alternating up- and downsampling (AUD) strategy to make the network focus not only on generating variants of HR features by using the upsampling layer, but also on projecting the generated HR features back into LR space using the downsampling layer. This can make better use of the spatial information and spectral information contained in HSI data.

B. Vision Transformer

Vision transformers [45], [46] have demonstrated their great potential ability in the computer vision community. They are widely used in various deep-learning models due to that the multihead self-attention (MSA) can model the long-range dependencies of image patches, such as image classification [47], object detection [48], and segmentation [49]. Although the vision transformers have shown their success in visual tasks, for visual SR tasks, the large computational cost usually hinders its practical application. To overcome this issue, the window-based MSA (W-MSA) of the swin transformer proposed by Liu et al. [50] reduces the computational complexity to be linear with respect to the image size. Zamir et al. [51] proposed multi-dconv head transposed attention (MDTA) for image SR, which calculates the self-attention across different channel dimensions, guaranteeing that the computational complexity does not increase quadratically with the increase of spatial resolution.

In this article, inspired by the self-attention mechanism of spatial and channel dimensions, and considering the spectral self-similarity of HSIs, we design a novel SSIT, which not only focuses on the spatial dimension, but also retains the spectral correlation between spectral bands. Moreover, the adaptive interaction of spatial and spectral feature information is performed via bidirectional interaction.

III. Our Proposed Method

In this section, we introduce the proposed AS^{3}ITransUNet in detail, including the overall network, SSIT block, multistage complementary information learning (MSCIL), and loss function.

A. Overall Network

Given the LR HSI, denoted by $\boldsymbol{I}_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B stand for the height, width, and spectral band number of HSIs, respectively. Instead of adopting the whole HSI as the input, we divide the LR HSI along the spectral dimension into nonoverlapping G groups as $\{L_1, \ldots, L_i, \ldots, L_G\}$, where $L_i \in \mathbb{R}^{H\times W\times m}$. Then each group is input to the proposed AS^{3}ITransUNet successively for processing. For the input group L_i, we map it to the feature map $\boldsymbol{X}_0 \in \mathbb{R}^{H\times W\times C}$ by 3×3 convolution, as shown in Fig. 1. The proposed AS^{3}ITransUNet has a U-Net structure, which includes an encoder and a decoder. Different from the existing U-Net framework, the proposed AS^{3}ITransUNet employs alternate upsampling and downsampling operations within the encoder–decoder mechanism, which greatly enhances the ability of the model to learn the mapping relationship between LR and HR. The input of the decoder is $\boldsymbol{X}_0$, after the first SSIT block, we adopt the upsampling operation to obtain the feature map $\boldsymbol{X}_1 \in \mathbb{R}^{rH\times rW\times C}$. To learn the complex mapping between the HR HSI and LR HSI, $\boldsymbol{X}_1$ passes through the second SSIT block and a downsampling operation to generate the feature $\boldsymbol{X}_2 \in \mathbb{R}^{H\times W\times 2C}$. At the end of the encoder, $\boldsymbol{X}_2$ is fed to the third SSIT block and processed by an upsampling operation to generate the feature $\boldsymbol{X}_3 \in \mathbb{R}^{rH\times rW\times C}$. The output of encoder $\boldsymbol{X}_3$ is fed to the bottleneck of U-Net which is the SSIT block.

The symmetric decoder is designed following the spirit of U-Net [52], [53] that combines the deep and shallow features of the corresponding stage through the skip connection (SC) to refine the features of HSIs. Unlike existing U-Net [52], [53], at each stage of the decoder, we not only connect the shallow features from the encoder with the decoder, but also effectively fuse the complementary information generated by the MSCIL. The output of the decoder feature map is operated

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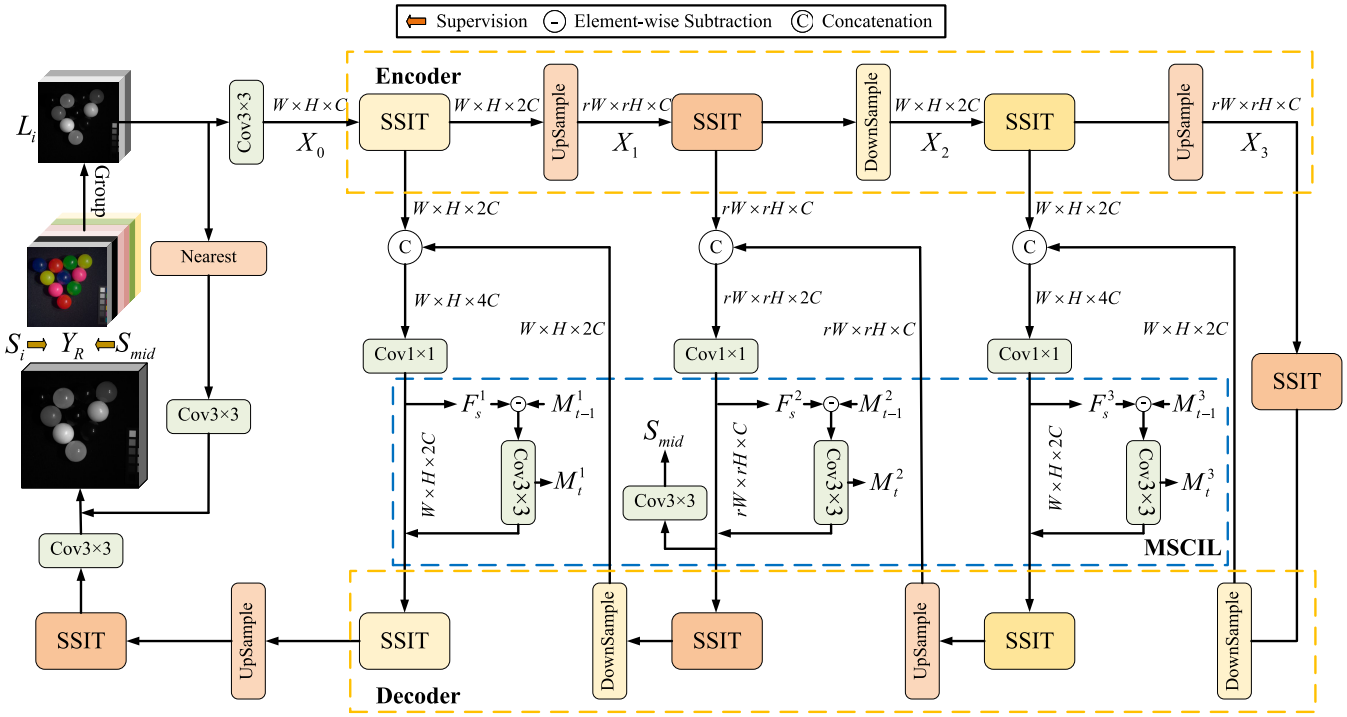


Fig. 1. Whole network architecture of the proposed method.

by upsampling. To further refine the feature, the SSIT block and a convolutional layer are applied.

Finally, as the input HSI group and the output HSI group have many similarities, we use the input HSI group for assisting the SR to generate the final SR HSI $\boldsymbol{S}_i \in \mathbb{R}^{rH\times rW\times \hat{m}}$, $(\hat{m} < m)$ by a cross-space residual composed of the nearest operation and a 3×3 convolution. To improve the stability and robustness of the proposed network, in addition to the loss of $\boldsymbol{S}_i$ and ground-truth $\boldsymbol{Y}_R$, the loss of the upsampled feature $\boldsymbol{S}_{\text{mid}}$ and ground-truth $\boldsymbol{Y}_R$ in the intermediate stage of the decoder is also used for generating a better feature map in the network.

B. SSIT Block

The CNN-based methods cannot model the correlation between the spectral bands of HSI effectively, whereas the strong correlation of HSIs is essential for the reconstruction and analysis of HSIs. The existing transformer models, although, can capture the long-range dependencies, directly using the self-attention on the spectral dimension for HSI SR may focus on some low-fidelity and less informative spatial regions. Therefore, to overcome the aforementioned problem, we propose the SSIT block, as shown in Fig. 2, which consists of the embedding, spatial branch, spectral branch, bidirectional interaction (spatial interaction unit and spectral interaction unit), and feed-forward network (FFN). Given the input feature map $\boldsymbol{X}_{\text{in}} \in \mathbb{R}^{H\times W\times \hat{C}}$, it is embedded by the embedding layer and divided into nonoverlapping local window features denoted by $\boldsymbol{X}_t \in \mathbb{R}^{(HW/M^2)\times(M\times M)\times \hat{C}}$, where $M \times M$ is the size of the window.

In the spatial branch, to prevent a significant increase in the amount of computation on spatial self-attention, the spectral dimension of the feature map is reduced by half through the linear layer and the feature map $\boldsymbol{X}_a \in \mathbb{R}^{(HW/M^2)\times(M\times M)\times(\hat{C}/2)}$ is obtained. Then, to extract the spatial information of the feature map, the local window self-attention (W-Attention) is used to model the long-range dependence between pixels on the spatial dimension. The feature map after the self-attention modeling is restored to the size of $H \times W \times (\hat{C}/2)$, which provides spatial information for spectral self-attention. The operation is represented as follows:

$$\hat{\boldsymbol{X}}_a = \text{Reverse}(\text{W-MSA}(\text{LN}(\boldsymbol{X}_a))) \tag{1}$$

where W-MSA$(\cdot)$ represents the self-attention operation [50] with local window, LN$(\cdot)$ represents the LayerNorm operation, and Reverse$(\cdot)$ represents the reverse operation of the feature map.

In the spectral branch, first, the input $\boldsymbol{X}_t$ passes through a linear layer and a layer norm. Then, the feature map $\boldsymbol{X}_e \in \mathbb{R}^{H\times W\times \hat{C}}$ is obtained by reverse operation. The aggregated spectral information is obtained by applying 1×1 convolution on $\boldsymbol{X}_e$. The 3×3 convolution is used to encode each spectral feature map into a token, yielding $\boldsymbol{Q} = W_d^Q W_p^Q \boldsymbol{X}_e$, $\boldsymbol{K} = W_d^K W_p^K \boldsymbol{X}_e$, and $\boldsymbol{V} = W_d^V W_p^V \boldsymbol{X}_e$, where $W_P^{(\cdot)}$ is 1×1 convolution and $W_d^{(\cdot)}$ is 3×3 depth-wise convolution. Next, $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are reshaped to generate $\hat{\boldsymbol{Q}} \in \mathbb{R}^{HW\times \hat{C}}$, $\hat{\boldsymbol{K}} \in \mathbb{R}^{\hat{C}\times HW}$, and $\hat{\boldsymbol{V}} \in \mathbb{R}^{HW\times \hat{C}}$, so that $\hat{\boldsymbol{Q}}$ and $\hat{\boldsymbol{K}}$ interact and generate a transposed attention map $\boldsymbol{Att}$ with the size of $\mathbb{R}^{\hat{C}\times\hat{C}}$.

For guiding spectral attention, we utilize valuable spatial information. Therefore, the spatial interaction unit is designed, as shown in Fig. 3(a), which generates the weights containing

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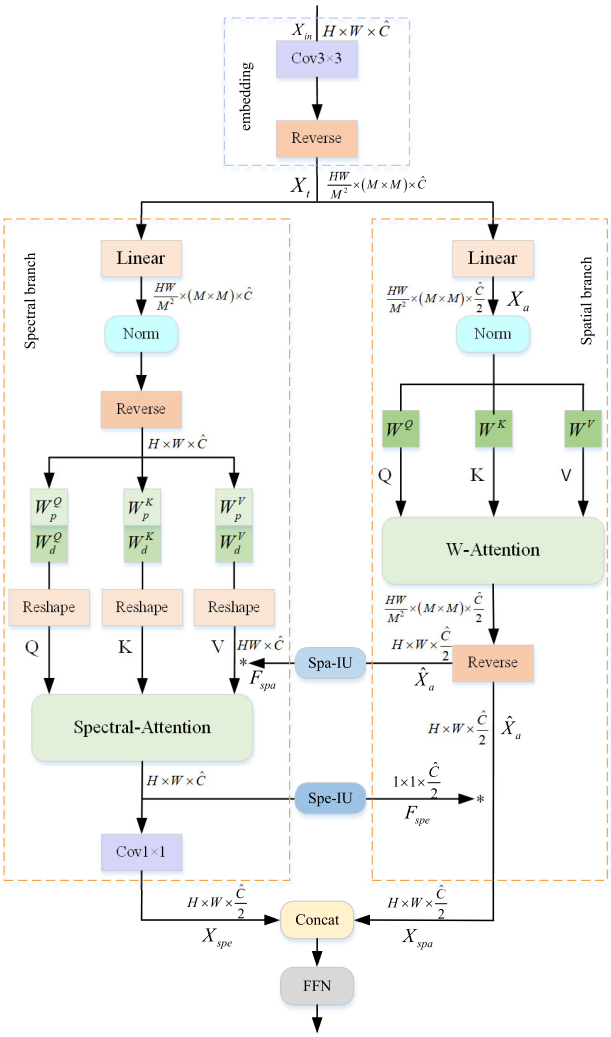


Fig. 2. SSIT block.

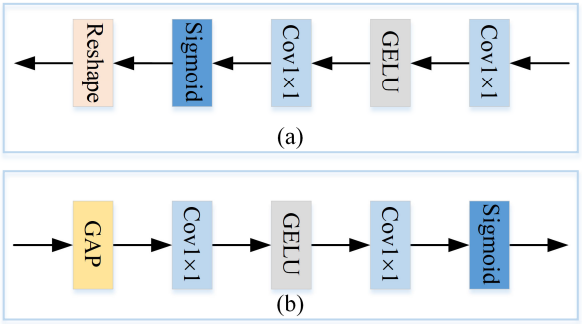


Fig. 3. Structure bidirectional interaction. (a) Spatial interaction unit (Spa-IU). (b) Spectral interaction unit (Spe-IU).

spatial information from the spatial branch. The interaction of spatial information can be expressed as

$$\boldsymbol{F}_{\text{spa}} = \text{reshape}(\text{Sig}(\text{Cov}(\text{GELU}(\text{Cov}(\hat{\boldsymbol{X}}_a))))) \tag{2}$$

where reshape$(\cdot)$ represents the reshape function, Sig$(\cdot)$ represents the sigmoid function, Cov$(\cdot)$ represents a 1×1 convolution operation, and $\boldsymbol{F}_{\text{spa}} \in \mathbb{R}^{HW \times \hat{C}}$. To guide the spectral self-attention branch with the spatial information, we reweighted $\hat{\boldsymbol{V}}$ with $\boldsymbol{F}_{\text{spa}}$ to guide it, which is calculated as

$$\hat{\boldsymbol{X}}_{\hat{e}} = (\boldsymbol{F}_{\text{spa}} * \hat{\boldsymbol{V}})\boldsymbol{Att}. \tag{3}$$

Then, $\hat{\boldsymbol{X}}_{\hat{e}} \in \mathbb{R}^{HW \times \hat{C}}$ is reshaped to obtain the original size $\hat{\boldsymbol{X}}_e \in \mathbb{R}^{H \times W \times \hat{C}}$.

For guiding the spatial branch, in the spectral interaction unit, as shown in Fig. 3(b), we follow the design of the SE layer [54]. The channel interaction consists of a global average pooling layer, followed by two successive 1×1 convolutional layers with activation between them (GELU [55]). Finally, sigmoid is used to generate the channel attention [54] in the spectral dimension which is to guide the spatial self-attention branch. The computation is expressed as follows:

$$\boldsymbol{F}_{\text{spe}} = \text{Sig}(\text{Cov}(\text{GELU}(\text{Cov}(\text{GAP}(\hat{\boldsymbol{X}}_e))))) \tag{4}$$

$$\boldsymbol{X}_{\text{spa}} = \hat{\boldsymbol{X}}_a * \boldsymbol{F}_{\text{spe}} \tag{5}$$

where GAP$(\cdot)$ represents the global average pooling operation, $\boldsymbol{F}_{\text{spe}} \in \mathbb{R}^{1 \times 1 \times (\hat{C}/2)}$, and $\boldsymbol{X}_a \in \mathbb{R}^{H \times W \times (\hat{C}/2)}$.

At the end of the SSIT module, the spectral dimension of $\hat{\boldsymbol{X}}_e$ is reduced by using 1×1 convolution and produces the output $\boldsymbol{X}_{\text{spe}}$. Finally, $\boldsymbol{X}_{\text{spe}}$ is concatenated with the output feature map $\boldsymbol{X}_{\text{spa}}$ of spatial branch, followed by the layer norm and multilayer perceptron (MLP) operation, the procedure is formulated as

$$\boldsymbol{X}' = \text{MLP}\big(\text{LN}\big(\text{Concat}\big(\boldsymbol{X}_{\text{spa}}, \boldsymbol{X}_{\text{spe}}\big)\big)\big) \tag{6}$$

where Concat$(\cdot)$ represents the concatenating operation. The proposed SSIT block can model the long-range dependencies both in the spectral and spatial dimensions and provide complementary clues in the spectral and spatial dimensions by the across-branch bidirectional interaction, thereby enhancing the feature representation ability of the network.

TABLE I
STUDY ON THE NUMBER OF GROUPS

Evaluation metric	1	3	5	7	9	11
PSNR	44.720	45.493	46.375	46.820	46.784	46.654
SSIM	0.9893	0.9903	0.9916	0.9919	0.9918	0.9917
SAM	2.843	2.688	2.420	2.364	2.362	2.363

TABLE II
STUDY ON THE WINDOW SIZE OF W-MSA IN SSIT

Evaluation metric	4×4	8×8	16×16	32×32
PSNR	44.720	46.820	46.798	46.644
SSIM	0.9893	0.9919	0.9919	0.9916
SAM	2.369	2.364	2.364	2.432

C. Multistage Complementary Information Learning

Due to the strong correlation between the adjacent HSI bands [25], it is reasonable to use the feature maps of the adjacent HSI group to assist the SR of the current HSI group which can further improve the HSI SR performance. To solve the limitation of existing one-stage fusion methods, we propose the MSCIL, as shown in Fig. 1. In our MSCIL, we design

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TABLE III
NETWORK STRUCTURE ANALYSIS

Variant	PSNR	SSIM	SAM
Ours w/o AUD	46.707	0.9919	2.389
Ours w/o SC	46.255	0.9910	2.542
Ours w/o MSCIL	46.235	0.9915	2.480
SSIT block is replaced with convolution	43.324	0.9879	2.983
Ours	46.820	0.9919	2.364

TABLE IV
STUDY ON THE ABLATION OF DIFFERENT COMPONENTS OF THE SSIT BLOCK

Commponet	The combination of different components					
Spatial branch	✓	×	✓	✓	✓	✓
Spectral branch	×	✓	✓	✓	✓	✓
Spa-IU	×	×	×	✓	×	✓
spe-IU	×	×	×	×	✓	✓
PSNR	46.590	45.151	46.307	46.603	46.363	46.820
SSIM	0.9915	0.9914	0.9915	0.9914	0.9918	0.9919
SAM	2.538	2.375	2.450	2.428	2.373	2.364

the multistage fusion for complementary information, which is obtained at different layer depths of the decoder. For the ith stage, let $\boldsymbol{M}_{t-1}^{i}$ and $\boldsymbol{F}_{s}^{i}$ represent the feature maps of the previous and current groups, respectively. Specifically, MSCIL receives $\boldsymbol{F}_{s}^{i}$ and $\boldsymbol{M}_{t-1}^{i}$, and the complementary information in the current group is learned as follows:

$$\boldsymbol{e}_{t}^{i} = \mathrm{Cov}_{3\times3}(\boldsymbol{F}_{s}^{i} - \boldsymbol{M}_{t-1}^{i}) \tag{7}$$

where $\mathrm{Cov}_{3\times3}(\cdot)$ denotes 3×3 convolution operation. Then, the complementary information is added to the current group feature $\boldsymbol{F}_{s}^{i}$ to obtain the fused output $\boldsymbol{F}_{d}^{i}$ as follows:

$$\boldsymbol{F}_{d}^{i} = \boldsymbol{e}_{t}^{i} + \boldsymbol{F}_{s}^{i}. \tag{8}$$

For learning the complementary information of the next group, $\boldsymbol{F}_{d}^{i}$ is then assigned to $\boldsymbol{M}_{t}^{i}$, that is,

$$\boldsymbol{M}_{t}^{i} = \boldsymbol{F}_{d}^{i}. \tag{9}$$

In the multiple stages, it should be noted that the resolutions of features are different. In the third stage, the resolution of $\boldsymbol{F}_{s}^{3}$ and $\boldsymbol{M}_{t-1}^{3}$ is $W \times H \times 2C$. After upsampling, the resolution of $\boldsymbol{F}_{s}^{2}$ and $\boldsymbol{M}_{t-1}^{2}$ is $rW \times rH \times C$ in the second stage. By using the downsampling, the resolution of $\boldsymbol{F}_{s}^{1}$ and $\boldsymbol{M}_{t-1}^{1}$ is $W \times H \times 2C$ in the first stage. Therefore, the HR and LR complementary information between the adjacent groups is learned in a multistage manner, which fuses the context information between adjacent groups effectively.

Note that when the number of bands within each group is large, the spectral curves between adjacent groups may be different. The missing complementary information in the current group cannot be effectively learned. When the number of bands in a group is small, the missing complementary information in the current group may not be only in the adjacent group. Thus, the number of bands in a group should be set appropriately and then the missing details in a specific band can be supplemented by the adjacent bands. The appropriate number of bands in a group is studied in Section IV-D. The proposed MSCIL learns complementary information with different resolutions in multiple stages, which can improve the reconstruction quality of HSIs.

D. Loss Function

During the training process of the network, each upsampling is optimized to generate an HR feature map. Therefore, we not only supervise the final output result, but also supervise the HR feature map of the intermediate process based on ground truth. In summary, the total loss L_{all} of the proposed method is defined as follows:

$$\boldsymbol{S}_{\mathrm{mid}} = \mathrm{Cov}_{3\times3}\left(\boldsymbol{F}_{d}^{2}\right) \tag{10}$$

$$L_{1} = \mathrm{MSE}(\boldsymbol{S}_{\mathrm{mid}}, \boldsymbol{Y}_{R}) \tag{11}$$

$$L_{2} = \mathrm{MSE}(\boldsymbol{S}_{i}, \boldsymbol{Y}_{R}) \tag{12}$$

$$L_{\mathrm{all}} = \lambda L_{1} + (1-\lambda)L_{2} \tag{13}$$

where $\boldsymbol{S}_{\mathrm{mid}}$ is the HR feature map of the second stage of the MSCIL, $\mathrm{MSE}(\cdot)$ is the MSE loss function, $\boldsymbol{F}_{d}^{2}$ represents the characteristics of the decoder output in the second stage, $\boldsymbol{Y}_{2}$ represents the final output of the network, $\boldsymbol{Y}_{R}$ represents the ground truth, and λ is a balanced hyperparameter.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, the three public datasets used in our experiments are first introduced. Next, the implementation setting and details are described. Then, we analyze the effectiveness of the proposed model. Finally, the quantitative and qualitative comparisons of the proposed method with seven state-of-the-art methods are presented. The peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and SAM are the quantitative indicators to evaluate the image quality after SR.

A. Datasets

1) CAVE: The CAVE dataset [56] is a popular natural HSI dataset collected by a cooled CCD camera at 10-nm intervals of 400–700-nm bands, containing 32 pairs of HSIs, corresponding to RGB images. The dimension of each image is $512 \times 512 \times 31$. The subjects include food, paint, toys, clothes, and faces.

2) Harvard: The Harvard dataset [57] contains 50 indoor and outdoor HSIs under daylight illumination and 27 images under artificial or mixed illumination. The spatial size of each HSI is 1024×1392, with 31 bands ranging from 420 to 720 nm.

3) Pavia Center: The Pavia Center dataset [58] was obtained by the ROSIS sensor. It contains only one HSI. After removing the noise bands, the number of bands of 1096×1096 pixels in the image is 102. Discard the area where the central part of the image does not contain information, leaving an information area with a space size of 1096×715.

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TABLE V
AVERAGE PSNR/SSIM/SAM OF DIFFERENT SCALE FACTORS ON THE CAVE DATASET QUANTITATIVELY COMPARES DIFFERENT METHODS, AND THE BEST AND SUBOPTIMAL PERFORMANCE ARE REPRESENTED BY BOLD AND UNDERLINE, RESPECTIVELY

scale factor	Evalution metric	Bicubic	GDRRN	3D-FCNN	SSPSR	NGswin	MCNET	RFSR	Ours
×2	PSNR ↑	41.924	42.936	44.655	-	43.116	<u>46.635</u>	45.602	**46.820**
	SSIM ↑	0.9856	0.9874	0.9894	-	0.9871	<u>0.9908</u>	0.9901	**0.9919**
	SAM ↓	3.011	2.984	2.361	-	3.404	<u>2.566</u>	2.688	**2.364**
×3	PSNR ↑	37.811	38.183	40.280	-	40.466	<u>42.910</u>	42.473	**42.958**
	SSIM ↑	0.9707	0.9734	0.9788	-	0.9792	<u>0.9844</u>	0.9839	**0.9846**
	SAM ↓	3.742	3.671	2.653	-	3.676	<u>2.902</u>	3.064	**2.886**
×4	PSNR ↑	35.747	36.229	37.533	39.759	37.851	<u>40.237</u>	40.199	**40.668**
	SSIM ↑	0.9550	0.9583	0.9660	0.9758	0.9667	<u>0.9777</u>	0.9775	**0.9783**
	SAM ↓	4.176	4.182	3.138	3.407	4.402	3.258	<u>3.251</u>	**3.191**

B. Implementation Details

All the datasets are first divided into the training and testing sets. For the CAVE and Harvard datasets, we randomly selected 80% of the samples from each dataset for training and randomly selected 24 patches from each image for enhancement. The testing samples are the remaining samples. For the Pavia Center dataset, 108 patches are selected from the left 1096 × 490 pixels as the training samples and the right part is cut into five images of 224 × 224 size as testing samples. All the training samples are enhanced by a scale factor of 1, 0.75, and 0.5, rotated by 90°, and flipped horizontally, respectively. Finally, these enhanced patches are downsampled by bicubic interpolation to obtain an LR HSI with the size of $B \times 32 \times 32$, where B is the total number of spectral bands.

For CAVE and Harvard datasets, both containing 31 spectral bands, the number of groups G is set to 7. For the Pavia Center dataset that has 102 spectral bands, the number of groups G is set to 17. The window size of W-MSA in SSIT is set to 8 × 8. During the training process, when the scale factor is 2 and 4, the number of filters is set to 48, and when the scale factor is 3, the number of filters is set to 36. The L_1 loss function is used for supervision, where λ in the loss function is 0.3. We use ADAM ($\beta_1 = 0.9$ and $\beta_2 = 0.999$) to optimize the proposed network. The initial learning rate is set to 10^{-4}, halving every 35 epochs. All the experiments are implemented in the PyTorch framework and conducted on an Intel Core i9-12900 KF CPU and two GTX3090 GPUs with 24 GB memory.

C. Ablation Study

To analyze the hyperparameters and study the proposed model, in this subsection, the ablation experiments are conducted on the CAVE dataset.

1) Study on the Number of Groups: To evaluate the effect of grouping for SR, we vary the number of groups to 1, 3, 5, 7, 9, and 11. From Table I, we can see that the performance of SR is affected by grouping. When there is no grouping, that is, the number of groups $G = 1$, the performance is the worst. This is because the spectral dimension of HSI data is high, the feature cannot be effectively extracted as all the spectral bands are processed together. When a large amount of spectral bands are divided into groups, each group is processed by the same number of filters, which facilitates the feature extraction. This is illustrated in Table I, and the SR performance becomes better as the number of groups G increases. But when the number of groups is greater than 7, the performance of the network begins to decline. This is because the spectral bands of each group are very few that the network cannot obtain more useful information and complementary information of adjacent groups. Therefore, we set the number of groups to 7 for the CAVE dataset. For the same reason, for the Pavia Center dataset, the number of groups is set to 17. It can be concluded that, under the condition of the same number of filter banks, it is recommended to divide the spectral bands into groups for effective feature extraction.

2) Study on the Window Size of W-MSA in SSIT: To show the impact of different window sizes of W-MSA, we set the window size to 4 × 4, 8 × 8, 16 × 16, and 32 × 32, and the results are shown in Table II. From the table, we can see that when the window size is 8 × 8, the performance is the best. When the window size is small, that is, 4 × 4, the PSNR and SSIM indices are low, because the long-range dependencies of the spatial dimension cannot be learned in the small window. When the window size is larger than 8 × 8, the performance of the model declines. The reason is that a large window will make the model receive more irrelevant information or noise, which leads to performance degradation. Therefore, we empirically select the window size to 8 × 8 for the W-MSA module in the SSIT.

3) Effectiveness of U-Net With Alternating Sampling: To verify that the adopted AUD strategy is more effective than the existing single upsampling strategy for HSI SR, we remove the up-down sampling operation in the AS^{3}ITransUNet and obtain a variant of our method. In Table III, we show the performance of the variant of this method, namely "ours w/o AUD." We can see that when the alternating sampling

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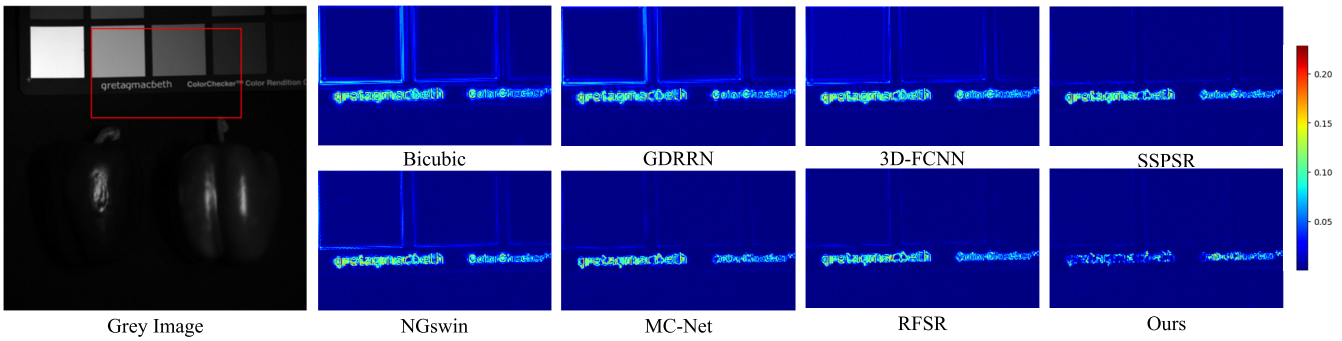


Fig. 4. Absolute error map comparisons for image *real_and_fake_peppers_ms* at 500 nm on the CAVE dataset with the scale factor ×4.

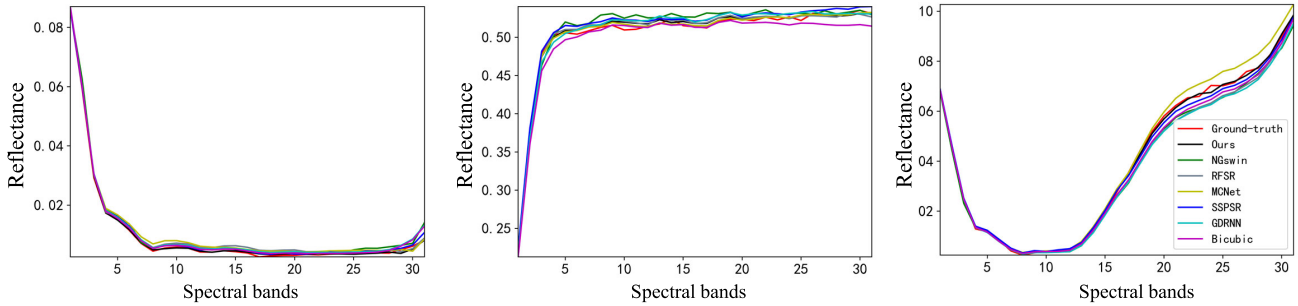


Fig. 5. Visual comparison of spectral distortion at pixel position (20, 20), (118, 118), and (288, 288) on the CAVE dataset with the scale factor ×4.

strategy is removed, the PSNR and SSIM indicators of the variant network decrease and the SAM indicator increases. Similarly, we remove the SC in the AS^{3}ITransUNet, that is, "ours w/o SC," and the performance is shown in Table III. We can observe that the evaluation indicators have a significant decline. Compared with the "Ours w/o SC" method that removes SCs, our method can make full use of the detailed information of HSIs by using the SC of the U-Net structure to combine shallow and deep hierarchical information. This verifies the effectiveness of the U-Net structure.

4) Effectiveness of SSIT: To verify the effectiveness of the SSIT block, we remove the SSIT block and replace it with a 3×3 convolution. As shown in Table III, when the SSIT block is replaced with convolution, there is a significant decrease in each evaluation index compared with the AS^{3}ITransUNet. This is mainly because after replacing the SSIT block, not only the correlation between bands is ignored, but also the long-range dependence modeling of spatial features is ignored.

5) Study on the Variant of SSIT: To further verify the effectiveness of bidirectional information interaction, spatial feature extraction, and spectral fidelity in the SSIT block, we conducted ablation experiments of bidirectional interaction, spectral branch, and spatial branch. The experimental results are shown in Table IV. First, when the SSIT block only has a spatial branch or a spectral branch, the evaluation indices are worse than those of SSIT. If the SSIT block lacks the spectral branch, the spectral fidelity ability decreases. If the SSIT block lacks the spatial branch, the spatial dimension information extraction ability decreases. When the SSIT block both has spatial and spectral branches, as we connect the halves of the spectral branch and the spatial branch, the SAM index is improved compared with the single spatial branch, and the PSNR and SSIM indices are improved compared with the single spectral branch. Then, if the spectral interaction unit is added, the spatial branch is guided by the spectral information, which leads to the decrease of SAM value and improves the spectral fidelity. If the spatial interaction unit is added, the spectral branch is guided by the spatial information, which leads to the increase of PSNR and SSIM values and the enhancement of spatial dimension performance. Then, when the spatial interaction unit and the spectral interaction unit are both added, the evaluation index value achieves the best effect, which verifies that the bidirectional interaction is an indispensable part of improving performance. Through the study on the variant of SSIT, we can find that each component is effective and improves the SR performance.

6) Effectiveness of MSCIL: To verify the effectiveness of our MSCIL, we conducted an ablation study on the proposed multistage fusion. The intergroup information fusion is performed at the end of the network like the one-stage methods. As shown in Table III, "Ours w/o MSCIL," that is, the multistage fusion strategy is removed, and it cannot achieve the best results, which demonstrates the effectiveness of our multistage fusion.

D. Comparisons With the SOTA Methods

To verify the performance of the proposed AS^{3}ITransUNet, we conducted comparative experiments in comparison with seven SOTA methods on three public hyperspectral datasets with 2, 3, and 4 scale factors, respectively. The compared methods include Bicubic, GDRRN [25], 3D-FCNN [28], SSPSR [26], NGswin [11], MC-Net [30], and RFSR [27]. It is

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TABLE VI

AVERAGE PSNR/SSIM/SAM OF DIFFERENT SCALE FACTORS ON THE HARVARD DATASET QUANTITATIVELY COMPARES DIFFERENT METHODS, AND THE BEST AND SUBOPTIMAL PERFORMANCE ARE REPRESENTED BY BOLD AND UNDERLINE, RESPECTIVELY

scale factor	Evalution metric	Bicubic	GDRRN	3D-FCNN	SSPSR	NGswin	MCNET	RFSR	Ours
×2	PSNR ↑	44.778	45.536	45.698	-	45.735	47.497	<u>47.517</u>	**47.615**
	SSIM ↑	0.9784	0.9812	0.9808	-	0.9815	0.9863	<u>0.9865</u>	**0.9869**
	SAM ↓	2.008	2.001	1.918	-	2.251	<u>1.874</u>	1.895	**1.867**
×3	PSNR ↑	41.748	42.294	42.320	-	43.313	<u>44.534</u>	44.491	**44.647**
	SSIM ↑	0.9619	0.9658	0.9646	-	0.9706	0.9752	<u>0.9753</u>	**0.9762**
	SAM ↓	2.277	2.264	2.191	-	2.358	<u>2.147</u>	2.191	**2.143**
×4	PSNR ↑	39.542	40.044	39.721	41.929	40.754	<u>42.285</u>	42.146	**42.435**
	SSIM ↑	0.9449	0.9496	0.9451	0.9600	0.9523	0.9621	<u>0.9633</u>	**0.9642**
	SAM ↓	2.471	2.469	2.386	2.344	2.605	<u>2.340</u>	2.400	**2.331**

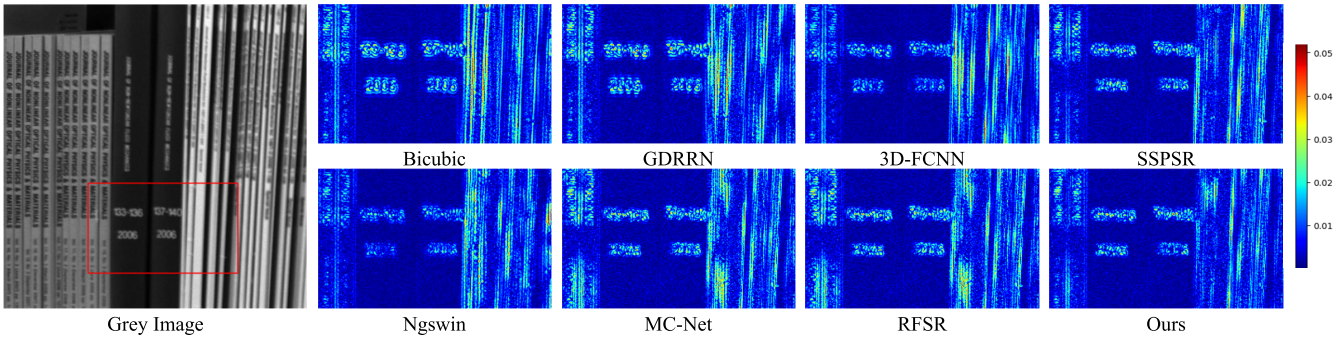


Fig. 6. Absolute error map comparisons for image *imgh*0 at 540 nm on the Harvard dataset with the scale factor ×4.

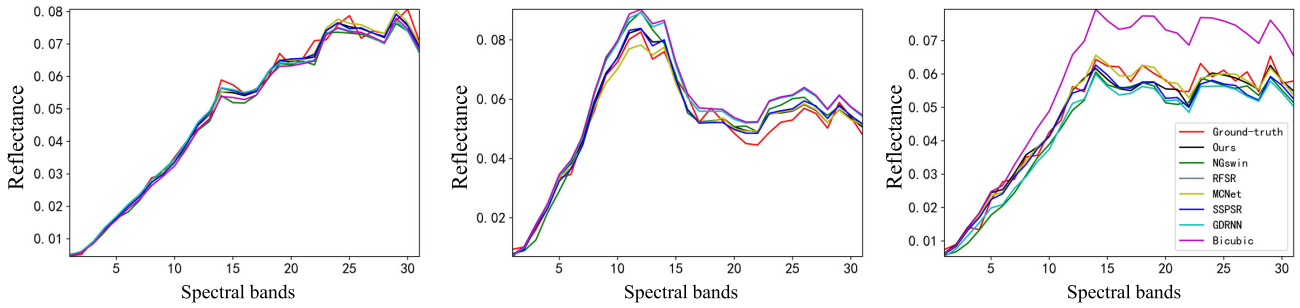


Fig. 7. Visual comparison of spectral distortion at pixel position (45, 45), (118, 118), and (300, 300) on the Harvard dataset with the scale factor ×4.

worth noting that the filling of 3D-FCNN is 0, which means that the output size of the network by 3D-FCNN has changed.

1) Results on the CAVE Dataset: Table V shows the quantitative comparison results of the eight methods on the CAVE dataset. We bold and underline the first and second best results. From the table, we can see that the proposed method outperforms the compared methods on three scale factors. The proposed method performs better when the scale factor is large. This might be attributed to the AUD strategy, which enables the network to learn more complex mappings between LR and HR. Compared with other methods, the SAM value of the proposed method is the lowest among all scaling factors. The possible reason is that our spectral branch models the spectral correlation and the spatial branch guides the spectral branch to focus on regions with high-fidelity spectral representation in the spatial dimension, which can better preserve the spectral features.

For the qualitative comparison of SR performance, the error maps of the reconstructed image and ground-truth image with one band produced by the eight methods on the CAVE dataset are shown in Fig. 4. The error maps show the reconstruction accuracy of each pixel from the aspect of spatial reconstruction quality of the image. The bluer the image in the absolute error map, the performance of the spatial reconstruction is better. We can see that the proposed AS^{3}ITransUNet has the lowest reconstruction error in comparison with other methods

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TABLE VII

AVERAGE PSNR/SSIM/SAM OF DIFFERENT SCALE FACTORS ON THE PAVIA CENTER DATASET QUANTITATIVELY COMPARES DIFFERENT METHODS, AND THE BEST AND SUBOPTIMAL PERFORMANCE ARE REPRESENTED BY BOLD AND UNDERLINE, RESPECTIVELY

scale factor	Evalution metric	Bicubic	GDRRN	3D-FCNN	SSPSR	NGswin	MCNET	RFSR	Ours
×2	PSNR ↑	34.306	34.936	36.174	-	35.518	36.689	36.694	**36.891**
	SSIM ↑	0.9286	0.9382	0.9538	-	0.9427	0.9549	0.9540	**0.9564**
	SAM ↓	3.954	4.010	3.031	-	4.224	3.573	3.654	**3.568**
×3	PSNR ↑	31.075	31.701	32.180	-	31.991	32.524	32.561	**32.694**
	SSIM ↑	0.8561	0.8750	0.8875	-	0.8809	0.8938	0.8943	**0.8988**
	SAM ↓	4.780	4.764	3.875	-	5.074	4.506	4.601	**4.490**
×4	PSNR ↑	29.296	29.785	30.090	30.492	29.917	30.357	30.489	**31.516**
	SSIM ↑	0.7929	0.8134	0.8246	0.8367	0.8210	0.8336	0.8368	**0.8471**
	SAM ↓	5.393	5.362	4.522	5.196	5.545	5.116	5.186	**5.113**

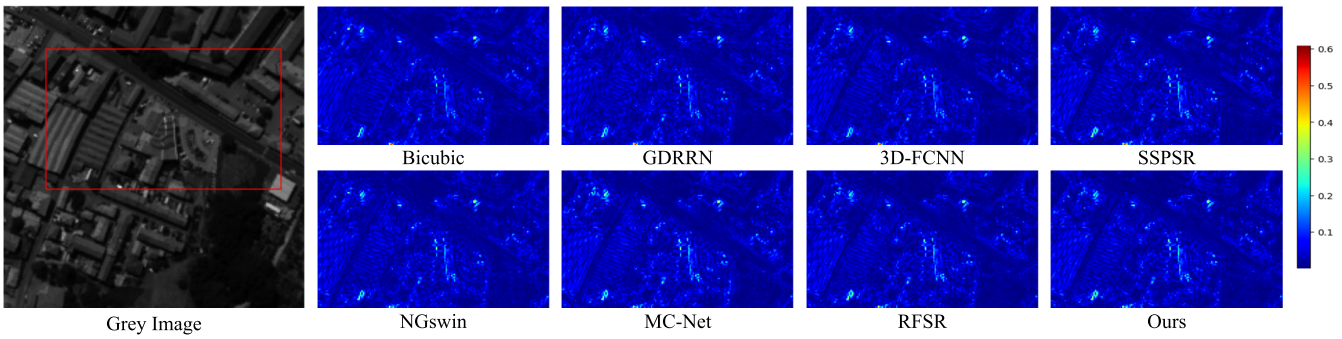


Fig. 8. Absolute error map comparisons for the 50th band on the Pavia Center dataset with the scale factor ×4.

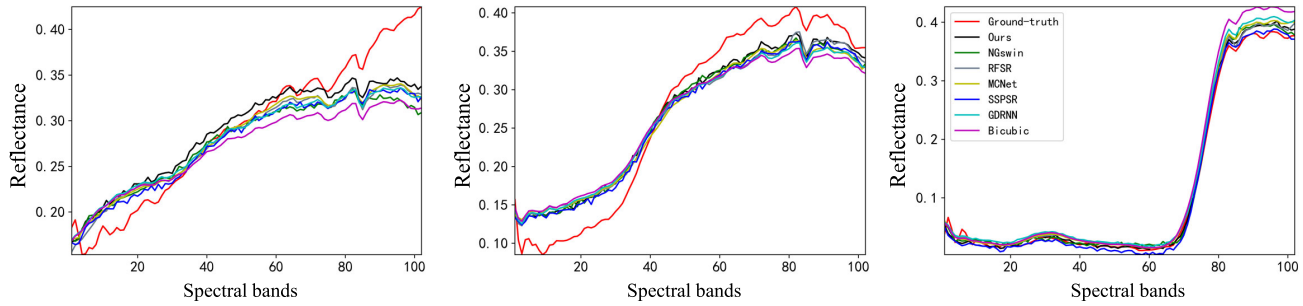


Fig. 9. Visual comparison of spectral distortion at pixel positions (22, 22), (99, 99), and (190, 150) on the Pavia Center dataset with the scale factor ×4.

and produces very shallow or no edges in the font structure and border areas, while other algorithms cannot recover the details of these areas, showing obvious texture information in these areas. To visualize the performance of various methods in terms of spectral fidelity, as shown in Fig. 5, we select and visualize the spectral values of three pixels. It can be found that the spectral value of our method is most closely matched with the ground-truth spectral value.

2) Results on the Harvard Dataset: Table VI reports the quantitative comparison results with eight HSI SR methods on the Harvard dataset. From the table, we can see that the proposed AS^{3}ITransUNet achieves the best comprehensive performance. The GDRRN that is based on 2-D convolution cannot fully extract the spectral features, resulting in poor results. Compared to the GDRRN, the 3D-FCNN improves the performance as the 3-D convolution can simultaneously extract spatial and spectral information, and the SSPSR also improves the performance as it uses 2-D convolution to extract spatial information and spectral attention to explore the correlation between HSI adjacent spectral bands. The MCNet that combines 2-D and 3-D convolution further explores the spatial information and extracts spectral information jointly. Therefore, the performance has been significantly improved. The RFSR uses the feedback network to mine the complementary spatial information between HSI bands and has better spatial information reconstruction performance. The MCNET

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TABLE VIII
AVERAGE PSNR/SSIM/SAM OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLE RATIOS

Training sample ratios	Evalution metric	Bicubic	GDRRN	3D-FCNN	SSPSR	NGswin	MCNET	RFSR	Ours
70%	PSNR	40.007	41.128	42.303	-	40.464	44.959	44.603	45.072
	SSIM	0.9831	0.9856	0.9876	-	0.9829	0.9915	0.9912	0.9921
	SAM	2.843	2.809	2.218	-	3.626	2.216	2.330	2.160
75%	PSNR	41.108	42.118	43.836	-	41.900	46.06	46.188	46.312
	SSIM	0.9842	0.9863	0.9889	-	0.9837	0.9918	0.9916	0.9924
	SAM	2.919	2.894	2.249	-	3.590	2.262	2.418	2.261
80%	PSNR	41.924	42.936	44.655	-	43.116	46.635	45.602	46.820
	SSIM	0.9856	0.9874	0.9894	-	0.9871	0.9908	0.9901	0.9919
	SAM	3.011	2.984	2.361	-	3.404	2.566	2.688	2.364

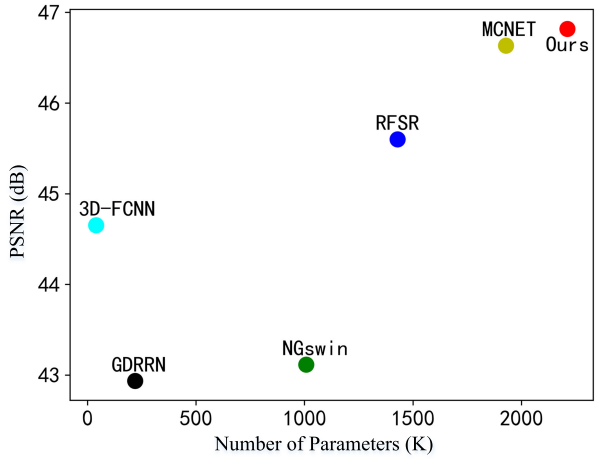


Fig. 10. Parameter studies on the CAVE dataset.

and RFSRNG methods provide comparable performance. The NGswin, which combines N-Gram context and transformer, enables the network to extend the receptive field by sliding the WSA to recover each degraded pixel. However, it fails to model the spectral correlation of HSIs, thus the SAM index is higher. The proposed AS^{3}ITransUNet can model the correlation between bands well and obtain a good SAM value.

In Fig. 6, we visualize the error maps of the reconstruction results of different competitive methods on the Harvard dataset with scale factor 4. It can be seen from Fig. 6, compared with other algorithms, the error value obtained by our method is closer to zero, especially the edge area of the book has a lower error value. It proves that our method can better restore the low-frequency and high-frequency detail information in the HSI. In addition, we show the spectral curves of three pixels of the test scene from the Harvard testing dataset in Fig. 7. It can be seen from the figure that compared with the spectral curves of different SOTA methods, the spectral curve value of the HSI reconstructed by our method is closer to the spectral curve value of the reference HSI.

3) Results on the Pavia Center Dataset: Table VII shows the quantitative performance of the Pavia Center dataset. From Table VII, we can note that the proposed AS^{3}ITransUNet gets the best performance in comparison with the SOTA methods. As the Pavia Center dataset has a large number of bands, there is more redundant information and complementary information between the adjacent groups. Our proposed method can make better use of the complementary information, thus obtaining better performance. Also, it produces quite promising performance in the visual comparison of absolute error maps and spectral distortions in Figs. 8 and 9.

E. Experiments on Different Training Sample Ratios

To evaluate the adaptability of the proposed method, we, respectively, select 70%, 75%, and 80% of the samples on the CAVE dataset as the training samples, and the remaining samples as the testing samples. The performance of AS^{3}ITransUNet and compared SOTA methods with different training sample ratios are tested on the sampling factor of 2, and the results are given in Table VIII. It can be seen from the figure that AS^{3}ItransUNet outperforms other methods using different proportions of training samples. It can be concluded that even with a limited number of training samples, the proposed AS^{3}ItransUNet shows better classification performance than other methods, which proves the adaptability and robustness of the proposed AS^{3}ItransUNet.

F. Computational Cost

To compare and analyze the computational complexity of the proposed AS^{3}ItransUNet, we show the model parameters and reconstruction PSNR value of all methods on the CAVE dataset with the scale factor 2 in Fig. 10. It can be observed that our method achieves the best performance with a slight increase in the amount of calculation.

V. CONCLUSION

This article proposes the SSIT U-Net with alternating sampling for HSI SR. We combine the alternating

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up- and downsampling method with U-Net, which effectively combines their advantages to enhance the flexibility of traditional encoder–decoder architecture for HSI SR. In the proposed SSIT block, a cross-branch bidirectional interaction is designed to achieve the adaptive interaction of spatial and spectral feature information. The MSCIL is proposed to supplement the details in the target band group by using the complementary information contained in the adjacent band group in multiple stages of the network. A large number of experiments on the widely used benchmark HSI datasets have proved the superiority of the proposed AS^{3}ITransUNet. In addition, sufficient ablation studies have been carried out, which demonstrate the effectiveness of each module in the proposed AS^{3}ITransUNet.

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